

PROFESSIONAL SUMMERY:

- Designed and implemented scalable data platforms on GCP, leveraging BigQuery for efficient data storage and processing. Developed sophisticated data platforms that enabled self-service analytics and real-time processing, supporting AI-driven decision-making across multiple business units.
- Architected cloud-native distributed data systems using GCP services, including Cloud Dataflow and Pub/Sub. Implemented robust data systems that handled petabytes of data, ensuring high availability and fault tolerance while optimizing for cost efficiency and sustainability.
- Developed complex ETL pipelines using advanced SQL and Python, processing large-scale datasets with a focus on performance. Created reusable SQL scripts and Python modules for data transformation, significantly reducing development time and improving code maintainability.
- Implemented real-time data streaming solutions using Kafka and Apache Beam, enabling near-instantaneous data processing. Optimized Kafka cluster configurations to handle high-throughput scenarios, ensuring data consistency and minimal latency in streaming pipelines.
- Built and maintained MLOps pipelines, integrating data engineering workflows with machine learning models. Implemented best practices in MLOps, including version control for models, automated testing, and continuous deployment, resulting in faster model iterations and improved prediction accuracy.
- Utilized Terraform and Jenkins for infrastructure-as-code and CI/CD automation of data pipeline deployments. Developed comprehensive Terraform modules for provisioning and managing GCP resources, while implementing Jenkins pipelines for automated testing and deployment of data engineering code.
- Designed and optimized data models and schemas using Erwin Data Modeler, ensuring efficient query performance and scalability. Applied best practices in data modeling, indexing, and partitioning to improve query execution times by orders of magnitude while maintaining data integrity.
- Implemented data governance and quality monitoring processes using Google Cloud Data Catalog and custom dashboards. Developed automated data validation checks and quality metrics, significantly reducing data inconsistencies and improving overall data reliability across the organization.
- Developed GraphQL APIs using Apollo GraphQL to expose complex data relationships and enable efficient data retrieval. Designed and implemented GraphQL schemas that simplified data access for front-end applications, reducing API call volume and improving application performance.
- Collaborated with data scientists to optimize data pipelines for machine learning models, improving model training efficiency. Implemented feature engineering pipelines using Apache Beam and Cloud Dataflow, reducing model training time and enhancing prediction accuracy for customer classification systems.
- Utilized Looker and Looker Studio to create interactive dashboards and visualizations for business intelligence. Developed custom Looker models and explores, enabling self-service analytics capabilities for non-technical users and driving data-informed decision-making across departments.
- Implemented advanced security measures in data platforms, ensuring compliance with data protection regulations. Developed role-based access control systems and encryption protocols for sensitive data, while maintaining audit logs for all data access and modifications.
- Designed and implemented data architecture for retail personalization systems, leveraging customer data and purchase history. Created scalable data models that supported real-time product recommendations and personalized marketing campaigns, significantly improving customer engagement and conversion rates.

- Developed custom ETL tools using Go and Java, optimizing performance for specific data processing requirements. Created efficient, concurrent data processing utilities in Go and Java, reducing execution time for complex ETL jobs by significant margins.
- Implemented knowledge graph solutions using Google Cloud Datastore to represent complex product and customer relationships. Designed flexible schema models in Datastore that captured intricate connections between products, categories, and customer preferences, enabling advanced query capabilities.
- Led cross-functional teams in Agile/DevOps environments, fostering collaboration between data engineers, scientists, and stakeholders. Conducted regular knowledge-sharing sessions on data engineering best practices, improving team efficiency and promoting a data-driven culture across the organization.
- Optimized data storage and processing costs on GCP by implementing intelligent data lifecycle management policies. Developed automated processes for data archiving and deletion, significantly reducing storage costs while maintaining compliance with data retention requirements.
- Contributed to the development of internal data processing frameworks, enhancing productivity across engineering teams. Created reusable libraries and tools for common data engineering tasks, reducing development time for new projects and promoting code standardization across the organization.

TECHNICAL SKILLS

- **Big Data Ecosystem Tools:** Apache Spark, Oozie, Zookeeper, Hadoop, Scala, Impala, Kafka, Nifi, Elastic MapReduce (EMR), Fact and Dimension Tables, Cloudera distribution, Hortonworks Ambari, HDFS, Map Reduce, YARN, PIG, Sqoop, HBase, Hive, Flink, Flume, Cassandra.
- **Relational and NOSQL Databases:** RDBMS, Oracle, SQL Server, PostgreSQL, DB2, DynamoDB, MongoDB, HBase, Cassandra, Cosmos DB.
- **Programming Languages:** SQL, PL/SQL, Python, Spark, JavaScript, HTML, SAS, Shell Scripting, Perl.
- **ETL and Data Integration:** IBM DataStage 11.5/9.1, Informatica, Talend 5.3/6.2, TOAD 10.5, Teradata SQL Assistant, Spring Boot, Databricks, Anaconda, Visual Source Safe, Putty, Notepad++, GitHub, Confluence, Jenkins.
- **Cloud Platforms:** Amazon Web Services (AWS), GCP, Azure.
- **Services:** Active Directory, Azure Monitoring, Azure Search, Data Factory, Key Vault, SQL Azure, Azure DevOps, Azure Synapse Analytics (DW), Azure Data Lake, AWS Lambda, Athena, AWS Elastic Compute Cloud (EC2), AWS Glue, Redshift, Kinesis, AWS Virtual Private Cloud (VPC), AWS CloudWatch, Step Functions, BigQuery, Composer, Google Cloud Storage, Cloud Dataproc.
- **Visualization and Reporting:** MS Excel, Power BI, Tableau, SPSS.
- **Containerization, Orchestration Tools:** Kubernetes, Docker, Docker Registry, Docker Hub.
- **Methodologies:** Agile, Scrum, Waterfall.
- **Libraries:** Pandas, NumPy, SciPy, Matplotlib, Seaborn, Scikit-Learn.
- **Algorithms:** Decision Tree, Random Forest, Logistic Regression, Gradient Boosting, Support Vector Machine (SVM), K-Nearest Neighbor (KNN).
- **IDE's & Frameworks:** Anaconda Software, Jupyter Notebook, Pycharm, Django, Flask.

PROFESSIONAL EXPERIENCE

Home Depot: Senior Data Engineer

March 2023 – Current

Location: Atlanta, GA

Environment: SQL, Python, Java, Scala, Go, BigQuery, GCP (Google Cloud Platform), Kafka, Apache Beam, Terraform, Jenkins, CI/CD, MLOps, Google Cloud Datastore, Cloud Dataflow, Pub/Sub, Looker, Looker Studio, Erwin Data Modeler, Google Cloud Data Catalog, Apollo GraphQL, AWS, Azure, ETL, Agile, DevOps, GraphQL, Git

- Designed and implemented a robust data pipeline using Apache Beam and Cloud Dataflow to process large-scale retail transaction data, improving data processing efficiency by optimizing Beam's parallel processing capabilities and Dataflow's auto-scaling features.
- Developed advanced ETL processes using Python and SQL to integrate multiple data sources, including point-of-sale systems and inventory management tools. Utilized Python's pandas library for data manipulation and SQL for complex joins and aggregations, enhancing data quality and consistency.
- Architected a real-time analytics platform using Kafka and BigQuery, enabling near-instantaneous insights into customer behavior and sales trends. Implemented Kafka streams for high-throughput message processing and leveraged BigQuery's streaming insert capabilities for real-time data ingestion.
- Led the migration of on-premises data warehouses to Google Cloud Platform (GCP), utilizing Terraform for infrastructure-as-code and Jenkins for CI/CD pipelines. Implemented GCP's best practices for security and compliance, ensuring data protection and regulatory adherence.
- Engineered a machine learning pipeline using MLOps principles, integrating model training, validation, and deployment workflows. Utilized TensorFlow and Scikit-learn libraries in Python to develop predictive models for customer churn and product recommendations.
- Implemented a data governance framework using Google Cloud Data Catalog and Erwin Data Modeler, enhancing data discoverability and metadata management. Utilized Data Catalog's automated tagging features and Erwin's data lineage capabilities to improve data traceability and compliance.
- Developed a GraphQL API using Apollo GraphQL and Go, providing a flexible interface for frontend applications to query retail analytics data. Leveraged Go's concurrency features and Apollo's caching mechanisms to optimize query performance and reduce latency.
- Designed and implemented a data lake solution using Google Cloud Storage and BigQuery, enabling cost-effective storage of historical data while maintaining query performance. Utilized BigQuery's external table functionality to seamlessly query data stored in Cloud Storage.
- Optimized data visualization workflows using Looker and Looker Studio, creating interactive dashboards for sales performance and inventory management. Leveraged Looker's LookML modeling language to define reusable dimensions and measures, improving report consistency and maintainability.
- Implemented a change data capture (CDC) system using Debezium and Kafka to capture real-time updates from relational databases. Utilized Kafka Connect for seamless integration with existing data pipelines, enabling real-time data replication and event-driven architectures.

Mastercard: Senior Data Engineer / Analyst
Location: Purchase, NY

May 2020 – February 2023

Environment: SQL, Python, Java, Scala, Go, BigQuery, GCP (Google Cloud Platform), Kafka, Apache Beam, Terraform, Jenkins, CI/CD, MLOps, Google Cloud Datastore, Cloud Dataflow, Pub/Sub, Looker, Looker Studio, Erwin Data Modeler, Google Cloud Data Catalog, Apollo GraphQL, AWS, Azure, ETL, Agile, DevOps, GraphQL, Git

- Designed and implemented complex data pipelines using Apache Beam and Cloud Dataflow to process high-volume banking transactions, improving data processing efficiency by optimizing Beam's windowing and triggering mechanisms for real-time analytics.
- Developed advanced machine learning models using Python and MLOps practices to predict customer churn and loan default risks, integrating model deployment pipelines with Jenkins for automated CI/CD, resulting in more accurate risk assessments.

- Created robust ETL processes using SQL and BigQuery to transform and load large-scale financial datasets, optimizing query performance through efficient table partitioning and clustering while ensuring data integrity and compliance with banking regulations.
- Implemented a real-time fraud detection system using Kafka and Java, processing streaming transaction data to identify suspicious patterns, while leveraging Kafka's exactly-once semantics to ensure no fraudulent activities were missed.
- Designed and maintained a data catalog using Google Cloud Data Catalog and Erwin Data Modeler, improving data discovery and governance across the organization, while integrating with BigQuery for seamless metadata management.
- Developed interactive dashboards using Looker and Looker Studio to visualize key banking metrics and KPIs, creating custom LookML models to ensure data consistency and enabling self-service analytics for business users.
- Implemented infrastructure as code using Terraform and GCP, automating the provisioning and management of cloud resources, while integrating with Jenkins for continuous deployment of infrastructure changes.
- Designed and developed a GraphQL API using Apollo GraphQL and Go, enabling efficient data fetching for front-end applications, while implementing robust authentication and authorization mechanisms to ensure data security.
- Optimized data storage and retrieval processes using Google Cloud Datastore and Cloud Dataflow, implementing efficient sharding strategies and leveraging Datastore's strong consistency model for critical financial transactions.
- Implemented a pub/sub system using Google Cloud Pub/Sub for event-driven architectures, enabling real-time updates for account balances and transaction notifications, while ensuring message delivery guarantees through Pub/Sub's at-least-once delivery semantics.

T-Mobile: Senior Big Data Engineer

August 2017 - April 2020

Location: Bellevue, WA

Environment SQL, Python, Java, Scala, Go, BigQuery, GCP (Google Cloud Platform), Kafka, Apache Beam, Terraform, Jenkins, CI/CD, MLOps, Google Cloud Datastore, Cloud Dataflow, Pub/Sub, Looker, Looker Studio, Erwin Data Modeler, Google Cloud Data Catalog, Apollo GraphQL, AWS, Azure, ETL, Agile, DevOps, GraphQL, Git

- Designed and implemented a real-time data processing pipeline using Apache Kafka and Apache Beam, enabling the ingestion and analysis of over 10 million telecom network events per hour. Utilized Kafka's partitioning and Beam's windowing features to optimize data throughput and processing efficiency.
- Developed and maintained complex ETL workflows in Google Cloud Dataflow using Python and Java, transforming raw telecom data into actionable insights. Leveraged Dataflow's auto-scaling capabilities and Python's data manipulation libraries to handle peak loads during network congestion periods.
- Created a machine learning pipeline for predictive maintenance of telecom infrastructure using MLOps practices and Google Cloud Platform services. Implemented version control and automated testing with Jenkins and CI/CD pipelines to ensure model reliability and reproducibility.
- Optimized BigQuery performance by redesigning table schemas and implementing partitioning and clustering strategies, resulting in faster query execution for large-scale telecom data analysis. Utilized BigQuery's cost estimation features to optimize resource allocation and reduce processing costs.
- Engineered a fault-tolerant data storage solution using Google Cloud Datastore and Cloud Dataflow, ensuring high availability of critical telecom customer data. Implemented Datastore's strong consistency model and Dataflow's exactly-once processing semantics to maintain data integrity during network outages.

- Developed a real-time network monitoring dashboard using Looker and Looker Studio, providing insights into network performance and customer experience. Leveraged Looker's LookML modeling language to create reusable data models for various telecom KPIs.
- Implemented a data governance framework using Google Cloud Data Catalog and Erwin Data Modeler, ensuring compliance with telecom industry regulations. Utilized Data Catalog's automated metadata discovery and Erwin's data lineage tracking to maintain data quality and traceability.
- Designed and deployed a scalable GraphQL API using Apollo GraphQL and Go, enabling efficient querying of telecom network data across multiple microservices. Leveraged Apollo's caching mechanisms and Go's concurrency features to handle high-volume API requests.
- Automated infrastructure provisioning and management using Terraform and Google Cloud Platform, ensuring consistent and repeatable deployments of telecom data processing systems. Implemented Terraform modules for managing GCP resources and utilized GCP's IAM policies for secure access control.
- Developed a streaming analytics solution using Google Cloud Pub/Sub and Cloud Dataflow to process and analyze real-time network traffic data. Utilized Pub/Sub's push delivery model and Dataflow's streaming engine to minimize latency in data processing and alerting.
- Implemented a data quality monitoring system using Apache Beam and BigQuery, ensuring the integrity of telecom billing and customer data. Leveraged Beam's data validation transforms and BigQuery's anomaly detection capabilities to identify and flag data inconsistencies.
- Optimized SQL queries and database schemas in BigQuery, improving query performance for large-scale telecom data analysis. Utilized SQL window functions and BigQuery's materialized views to enhance complex analytical queries on customer behavior and network usage patterns.
- Developed a custom ETL framework using Scala and Apache Beam, enabling flexible data transformation pipelines for various telecom data sources. Leveraged Scala's functional programming features and Beam's unified programming model to create reusable and composable data processing components.
- Implemented a secure data access layer using Google Cloud Datastore and Apollo GraphQL, ensuring fine-grained access control for sensitive telecom customer information. Utilized Datastore's strong consistency model and Apollo's authentication directives to enforce data privacy policies.
- Created a CI/CD pipeline using Jenkins and Google Cloud Platform services, automating the deployment and testing of big data applications in the telecom environment. Leveraged Jenkins' declarative pipeline syntax and GCP's container registry for efficient and reliable software delivery.

Southwest Airlines: Big Data Engineer

December 2015 - July 2017

Location: Dallas, TX

- Worked with Google Cloud Platform (GCP) services, including BigQuery, Google Cloud Storage (GCS) bucket, Google Cloud Functions, Cloud Dataflow, Dataproc, and Google Cloud operations.
- Utilized Cloud Shell SDK in GCP to configure services such as Dataproc, Storage, and BigQuery.
- Utilized GCP as a managed workflow service for orchestrating complex data workflows and pipelines.
- Implemented identity and access management (IAM) policies and security controls using Google Cloud IAM and Security Command Center to ensure data confidentiality, integrity, and availability.
- Implemented monitoring and logging solutions using Google Cloud Monitoring and Logging to track the health, performance, and availability of data pipelines and infrastructure components.
- Enhanced data pipelines utilizing tools such as GCP Dataflow, Apache Beam, or Apache Spark to handle the ingestion, processing, and transformation of extensive datasets from diverse origins.
- Utilized G-cloud function with Python to load data into BigQuery for on-arrival CSV files in GCS bucket and analyzed various types of raw files like JSON, CSV, and XML with Python using Pandas and NumPy.
- Utilized REST API with Python for ingesting data into BigQuery.

- Used Informatica to extract, transform, and load data from SQL Server to Oracle databases.
- Designed and created data cleansing, data conversion, validation, and external loading scripts like MLOAD and FLOAD for Teradata warehouse using Informatica ETL tool.
- Managed and optimized GCP infrastructure resources, including compute, storage, and networking, to enhance cost efficiency and scalability in data operations.
- Integrated Hadoop with other Big Data technologies, such as Hive and Pig, for advanced data analysis and visualization.
- Developed complex SQL queries and scheduled data extraction and transformation tasks using Cloud Composer, resulting in a 30% reduction in data processing time and improved data-driven decision-making across the organization.
- Utilized Hadoop Distributed File System (HDFS) for efficient storage and retrieval of massive data.
- Collaborated with data scientists to translate business requirements into Hadoop solutions.
- Possess expertise with relational database concepts, stored procedures, functions, triggers, and Scalability analysis.
- Optimized SQL and PL/SQL queries using different tuning techniques like using hints, parallel processing, and optimization rules.
- Integrated with cloud services to facilitate seamless data migration and synchronization.
- Developed high-throughput data ingestion pipelines using Kafka, Pub/Sub, and Cloud Dataflow, ensuring real-time processing at millisecond latency.
- Integrated Google Cloud Secret Manager with ETL pipelines, securing API keys and database credentials from unauthorized access
- Participated in user meetings, gathered business requirements & specifications for the Data-warehouse design. Translated the user inputs into ETL design docs.
- Collaborated on the integration of GCP services with existing tools and platforms across the organizational landscape, including BI tools, data analytics platforms, and third-party APIs, to facilitate smooth data integration and analysis processes.
- Designed multi-region BigQuery datasets to improve data locality and query response times.
- Developed streaming applications using PySpark to read data from Kafka and persist it in NOSQL databases such as HBase and Cassandra.
- Implemented cost-efficient storage tiering policies in Google Cloud Storage, transitioning cold data to Nearline and Archive storage for reduced expenses.
- Designed and implemented multi-region data replication strategies using BigQuery scheduled queries and cross-region datasets.
- Integrated Google Data Studio with BigQuery, building interactive dashboards that enhanced decision-making with real-time analytics.
- Automated schema evolution handling in BigQuery for dynamic data updates.
- Developed scripts to automate data movement between Cloud Storage and BigQuery.
- Configured Google Cloud Storage lifecycle policies for cost-effective archival and tiered storage management.
- Optimized message throughput and latency in Pub/Sub for handling high-velocity data streams.
- Developed parameterized Cloud Composer DAGs to automate ETL workflows, ensuring dynamic source-to-destination mapping across environments.
- Built custom logging and monitoring frameworks using Google Cloud Operations Suite, providing real-time alerts on pipeline failures and performance bottlenecks.
- Developed CI/CD pipelines in GitHub Actions and Cloud Build, automating ETL workflow deployments for Google Cloud Composer and Dataflow.
- Created serverless data processing solutions using Google Cloud Run and Cloud Tasks, improving cost efficiency for transient workloads.
- Optimized data processing workloads by implementing auto-scaling Compute Engine instances for ETL jobs.
- Implemented identity and access management (IAM) policies and security controls using Google Cloud IAM and Security Command Center to ensure data confidentiality, integrity, and availability.
- Developed monitoring scripts using BigQuery audit logs to track query execution and optimize resource usage.
- Building, publishing customized interactive reports and dashboards, and report scheduling using cu Server.
- Created action filters, parameters, and calculated sets for preparing dashboards in Tableau.
- Designed multi-region BigQuery datasets to improve data locality and query response times.
- Maintain scripts/job queues in sync with IT updates on Servers.
- Implement support structure for underlying DB query scripts, job queues, usage tracking, etc.

Environment: Informatica, Google Cloud Platform (GCP), Dataflow, Apache Beam, Oracle 11g, SQL Server 2008, ETL, Hadoop, PL/SQL, Cassandra, API, Hive, Kafka, Apache Spark, Dataproc, GCP Cloud Storage, Teradata, Visual Studio, IAM, BigQuery, JSON, Rest API, Pandas, NumPy, Tableau, MS Excel, MS Access.

Amazon India: ETL/Data Warehouse Developer
Location: Bangalore, India

June 2014 - October 2015

- Assisting the team with performance tuning for **ETL** and database processes.
- Developed and maintained Scalable and high-performance data warehouse solutions using technologies such as **AWS Redshift, Snowflake**.
- Developed dimensional models and schemas using techniques like star and **Snowflake** schemas for efficient data storage and retrieval.
- Developed data ingestion workflows using **AWS S3** for staging data files and **Redshift Spectrum** to query data directly from **S3**.
- Integrated data from diverse sources into **Amazon Redshift** using **AWS** services like **AWS Data Pipeline** or third-party **ETL** tools that supported **Redshift**.
- Developed data ingestion pipelines using **Informatica** workflows to streamline data flow into the data warehouse.
- Implemented security measures for **Amazon Redshift** using **AWS** Identity and Access Management (**IAM**) to control access to **Redshift** clusters and data.
- Performed database optimization tasks such as indexing and partitioning to enhance query performance and overall system efficiency.
- Monitored and scheduled ETL workflows using **AWS Data Pipeline** with dependency management.
- Created interactive dashboards and reports using **Power BI** connect to data sources in **Teradata**.
- Documented data warehouse design, **ETL** processes, and data models using tools like Microsoft Word or Confluence for effective knowledge sharing.
- Monitored and optimized **Redshift** cluster performance using **AWS CloudWatch** for metrics and alerts related to query performance and resource utilization.
- Integrated **AWS CloudWatch** logs with **AWS Lambda** to monitor and respond to ETL failures in real time.
- Developed real-time ETL processes by integrating Amazon Kinesis with S3 and Redshift for streaming data ingestion.
- Automated data ingestion pipelines using **AWS Data Pipeline** to move data across RDS, DynamoDB, and S3.
- Implemented error-handling and retry mechanisms for ETL workflows using **AWS Data Pipeline**.
- Conduct **ETL** performance tuning, troubleshooting, support, and capacity estimation.
- Worked extensively on different types of transformations like source qualifier, expression, filter, aggregator, rank, lookup, stored procedure, sequence generator and joiner.
- Developed shell scripts for automation of **Informatica** session loads.

Environment: Windows, AWS Redshift, Snowflake, IAM, Power BI, Confluence, CloudWatch, AWS, EC2, S3, ETL, SQL, Informatica 5.1, MS Excel.

Education:

RVR & JC College of Engineering,
Bachelor of Technology in Computer Science

June 2010 – May 2014